KARTHIKEYA MADALA

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EXPERIENCE

Technical Support - Systems Engineer, Texas State University | Austin, TX **_Aug 2025 – May 2026_**

- Developed scalable backend services in Python (FastAPI and Flask) to manage campus support tickets and inventory records, improving maintainability and accelerating high-volume data processing.
- Designed normalized schemas in PostgreSQL with SQLAlchemy ORM to model ticket lifecycles, asset tracking, and resource allocation, enabling faster queries and reducing inconsistency across modules.
- Implemented REST APIs in Python with GraphQL for secure role-based operations and real-time dashboards used by students, staff, and administrators.
- Integrated Apache Kafka with Python microservices, PostgreSQL, and Redis to provide real-time notifications, ETA updates, and reliable message delivery under concurrent loads.
- Deployed containerized services with Docker and Kubernetes on Azure, setting up CI/CD pipelines with GitHub Actions for seamless scaling and zero-downtime releases.

IT Analyst Intern, Texas Commision Environment Quality HQ | Austin, TX **_May 2025 – Aug 2025_**

- Engineered distributed workflows in Python and Scala to orchestrate emissions pipelines across Azure Data Lake, enabling scalable, fault-tolerant processing and reducing downtime during peak loads.
- Developed REST APIs in Python (FastAPI) to expose secure emissions data access, integrating with PostgreSQL and Kafka for high-throughput streaming ingestion.
- Automated validation frameworks in Python and SQL to ensure metadata accuracy and compliance consistency across cloud-hosted datasets, reducing redundant manual checks.
- Optimized reporting workflows by introducing Redis caching, partitioning, and indexing strategies in Azure, cutting runtime by 25% and enabling faster large-scale simulations.

Data Engineer Intern, *Unacademy* **_May 2023 – Aug 2023_**

- Built and optimized ETL pipelines in Apache Spark (Python and PySpark) orchestrated with Apache Airflow to process event data from Amazon S3, accelerating aggregation and reducing reporting latency.
- Implemented storage workflows in MongoDB and MySQL, designing schemas to model user activity, metadata, and relational mappings that supported real-time queries, ensured schema evolution, and improved consistency across distributed systems.
- Developed backend services in Python (Flask and FastAPI) with GraphQL APIs, incorporating modular architecture and role-based access control to enhance scalability, streamline integration with client applications, and improve overall system reliability.
- Optimized Spark jobs and SQL queries with partitioning, caching, and indexing, reducing runtimes by 30% and lowering infrastructure costs while collaborating through Git-based workflows (GitHub and Bitbucket).

PROJECTS

Online Exam Platform *(Python | Django | React.js | PostgreSQL | Redis | AWS | Docker | Kubernetes)*

- Designed microservices for authentication, scheduling, and ensuring secure workflows and maintainable code structure.
- Implemented session handling and scheduling algorithms, enabling concurrent users with minimal latency during peak sessions.
- Deployed containerized services on AWS with Redis caching, applying load balancing and fault tolerance to maintain availability.

Campus Ticketing & Inventory System *(Python | Flask | GraphQL | Kafka | Spark | PostgreSQL | MongoDB | Azure)*

- Built scalable Python microservices with GraphQL APIs to handle 50K+ ticketing and inventory requests efficiently.
- Integrated Kafka with Spark for real-time ETA predictions and concurrent request handling in distributed workloads.
- Modeled PostgreSQL and MongoDB databases, enabling fast queries, flexible schemas, and resilient storage.

SKILLS

Languages: Java, Python, SQL, HTML/CSS, JavaScript
Frameworks & Libraries: FastAPI, Flask, Django, React.js, GraphQL, PySpark, Apache Kafka, Apache Spark, Airflow, Pandas
Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Amazon S3, Azure Data Lake
Cloud Platforms: AWS (EC2, S3, RDS, Lambda, API Gateway), Azure (App Services, Data Factory), GCP (Cloud Run, BigQuery)
DevOps & CI/CD: Docker, Kubernetes, Git, Jenkins

EDUCATION

Master of Science, Data Analytics and Information Systems Aug 2024–May 2026
Texas State University | San Marcos, TX CGPA: 3.83/4
Coursework: Software Engineering, Advanced DBMS, Data Structures & Algorithms, Cloud Computing, Machine Learning
Bachelor of Technology, Computer Engineering Specialised in Machine Learning Aug 2020–Apr 2024
Gandhi Institute of Technology and Management (GITAM) CGPA: 3.25/4